

Amey Pasarkar

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EDUCATION

Princeton University

PhD candidate in Machine Learning, Advisor: Adji Bousso Dieng

Princeton, NJ
2022-2027

- Awarded 2022 NSF-GRFP Fellowship

Columbia University

Bachelor of Science in Operations Research

New York, NY
2018 – 2022

- Cumulative GPA: 4.02/4.00, Magna Cum Laude
- Finalist for Computing Research Association Undergraduate Research Award

EXPERIENCE

Google Cloud, TPU Inference

Software Engineering Intern

Sunnyvale, CA

June 2026 – Aug 2026

- Developed new inference algorithms for accelerating diffusion image generation models on Google's JAX + TPU stack, targeting lower-latency and higher-throughput deployment.
- Designed custom implementations of various popular text-to-image diffusion models, providing an over 25% speed-up over SGLang GPU baselines. All contribution made opensource on Google's [MaxDiffusion](#) repository.

Princeton Department of Computer Science, Vertaix

PhD Student

Princeton, NJ

March 2023 – Present

- Building dataset curation methods for billion-scale generative AI training corpora to identify redundancy, memorization, distributional gaps, and data quality issues.
- Developing data-efficient methods for LLM post-training and data acquisition, including coreset selection for fine-tuning and active search, to identify high-value examples while reducing data and compute requirements.
- Building trustworthy AI evaluation methods, including OOD and anomaly detection, to identify model failure modes, measure robustness, and distinguish genuine generalization from memorization.

Amazon, FireTV Science

Research Scientist Intern

Sunnyvale, CA

May 2025 – Aug 2025

- Trained LLMs to improve video search query understanding on the Amazon FireTV.
- Designed and implemented training and evaluation pipelines for LLMs ranging from 0.6 to 70 billion parameters.
- Applied Chain-of-Thought and Direct Preference Optimization to improve model performance, achieving 5× lower inference latency and 10% higher accuracy compared to the baseline production model.

Columbia Department of Computer Science, Itsik Pe'er Lab

Research Assistant

New York, NY

Aug 2019 – Dec 2021

- Developed probabilistic dynamical models for noisy, high-dimensional time-series and spatial datasets.
- Created new class of latent-variable mixture models to represent structured, spatially dependent distributions.
- Derived scalable approximate inference algorithms for these models by extending techniques like variational inference and expectation-maximization.

Amazon, AWS Safety Engineering

Software Development Engineer Intern

Seattle, WA

May 2021 – Aug 2021

- Built and deployed pipeline for performing context-based queries on documents detailing incidents in all Amazon services. Model results helped identify vulnerabilities in services and improve their reliability.

SELECTED PUBLICATIONS

- Amey P. Pasarkar**, Adji Bousso Dieng. “Vendi Novelty Score for Out-of-Distribution Detection.” *arXiv* preprint (under review), 2026.
- Amey P. Pasarkar**, Adji Bousso Dieng. “The Vendiscope: An Algorithmic Microscope for Data Collections.” *arXiv* preprint (under review), 2025.
- Amey P. Pasarkar**, Adji Bousso Dieng. “Cousins of the Vendi Score: A Family of Similarity-Based Diversity Metrics for Science and Machine Learning.” *AISTATS*, 2024.
- Amey P. Pasarkar**, Gianluca M. Bencomo, Simon Olsson, Adji Bousso Dieng. “Vendi sampling for molecular simulations: Diversity as a force for faster convergence and better exploration.” *Journal of Chemical Physics*, 2023.
- Amey P. Pasarkar**, Tyler A. Joseph, and Itsik Pe'er. “Directional Gaussian Mixture Models of the gut microbiome elucidate microbial spatial structure.” *mSystems*, 2021.